

BC1 Equal-Temperature Known-Answer Diagnostic Projection-Free Controlled Integrator

Numerical audit report

9 September 2026

1 Question and frozen decision rule

For the canonical BC1 bath at $T_1 = T_n = 6$, the stationary Gibbs density is invariant under phase reversal, while every $\sin \theta_j$ is odd. Therefore $\langle \sin \theta_j \rangle = 0$ exactly. The inherited SIMD calculation failed this control with maximum $|z| = 8.678$ at $n = 25$ and 12.929 at $n = 100$. The diagnostic repeats only BC1 with the controlled Cartesian implementation: double precision, fixed $dt = 5 \times 10^{-4}$, standard trigonometry, no adaptive stepping, and no positive-action projection or floor.

The gate was frozen before production. The port is accepted only if the spatial mean action is within 5% of the SIMD reference and its relative spatial range is no larger than the SIMD range. The sine control is assessed both pointwise at $|z| > 1.95996$ and simultaneously with a two-sided Bonferroni threshold.

2 Protocol and integrity

For $n = 25$, burn-in and measurement times were 2000 and 2000. For $n = 100$, they were 32000 and 2000. The sample interval was 0.01, producing 200001 profile samples per trajectory. Two base seeds and 256 trajectories per base seed gave 512 independent trajectories at each chain length.

n	base seeds	trajectories	samples/trajectory	integrity errors
25	2026090825, 2026090826	512	200001	0
100	2026090900, 2026090901	512	200001	0

All 426 frozen output hashes verified, and a fresh independent invocation of the frozen analysis reproduced the saved JSON byte-for-byte.

3 Primary result

n	controlled max $ z $	SIMD max $ z $	pointwise failures	Bonf. critical	simultaneous test
25	1.875	8.678	0/24	3.078	PASS
100	2.984	12.929	5/99	3.478	PASS

At $n = 25$, no bond fails even the uncorrected pointwise test. At $n = 100$, 5 of 99 bonds exceed 1.95996, approximately the number expected by chance under 99 pointwise 5% tests, while the maximum $|z| = 2.984$ remains below the predeclared simultaneous threshold 3.478. Thus the exact zero-sine control passes at both lengths under the controlled integrator.

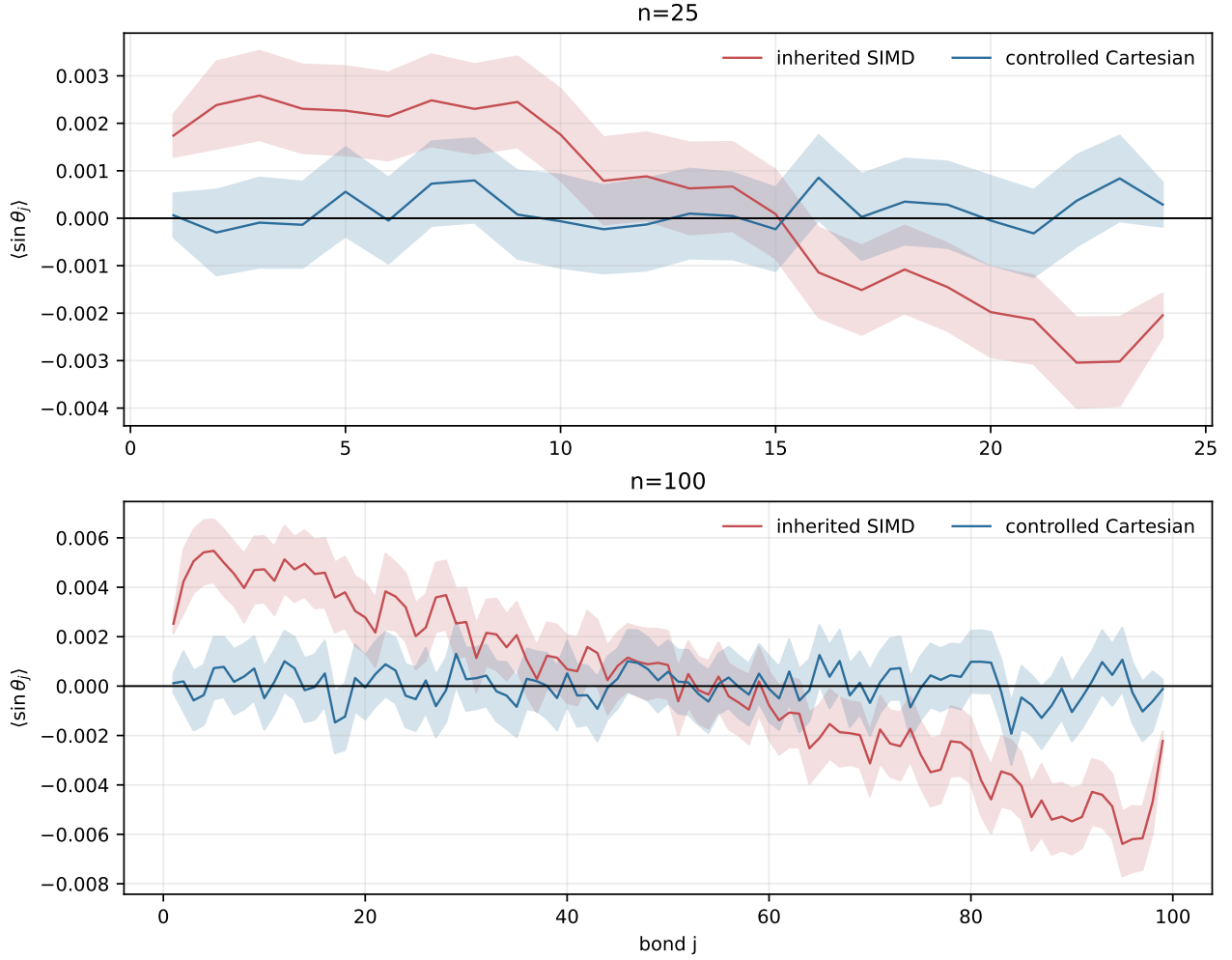


Figure 1: Bond-sine profiles and 95% pointwise bands. The inherited SIMD profile shows coherent nonzero structure; the controlled profile fluctuates around zero.

4 Like-for-like precision

The mean controlled-to-SIMD SE ratios are comparable or smaller: the median ratio is 0.979 at $n = 25$ and 0.931 at $n = 100$. Mean controlled SEs are 0.000450711 and 0.000638362, compared with SIMD means 0.000459151 and 0.000687433. The disappearance of the coherent signal is therefore not caused by reduced precision.

5 Porting gate: action profile

n	controlled mean	SIMD mean	relative difference	controlled range	SIMD range
25	0.496684	0.494670	0.407%	0.706%	0.894%
100	0.245984	0.249219	1.298%	1.725%	9.731%

Both predeclared porting gates pass. The spatial mean differs by only 0.407% at $n = 25$ and 1.298% at $n = 100$, and the controlled profiles are flatter than their SIMD references.

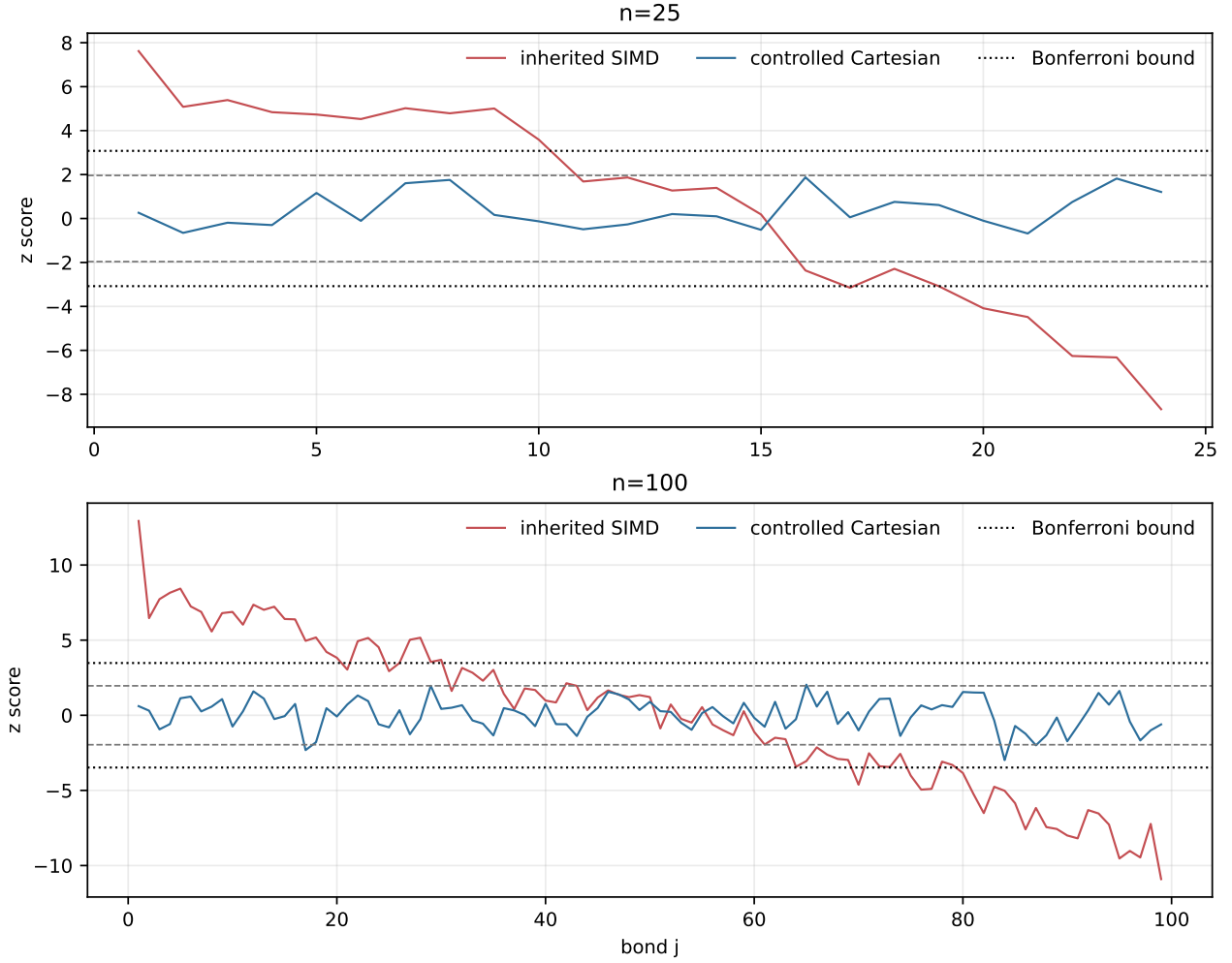


Figure 2: Per-bond z scores. Dotted lines are the frozen Bonferroni bounds; dashed lines mark the uncorrected pointwise 95% bounds.

6 Numerical-event audit

n	projections	floors	zero-radius	midpoint failures	min sampled action
25	0	0	0	0	$4.858175\text{e-}11$
100	0	0	0	0	$1.683034\text{e-}12$

Nonfinite batches were zero at both lengths. The near-zero-action region was sampled down to 4.86×10^{-11} and 1.68×10^{-12} without clipping.

7 Verdict and claim boundary

The controlled scheme passes the exact BC1 equal-temperature sine control at both chain lengths, whereas the inherited SIMD scheme fails systematically at comparable SE. Together with the passed action-profile porting gate, this establishes that the old nonzero equal-temperature bond-sine profile is a numerical artefact of the inherited SIMD numerical scheme, not a property of the BC1 stationary dynamics. Because several numerical changes were made together, this experiment does not identify projection alone as the unique cause.

Consequently, driven $(10, 2)$ bond-sine profiles obtained with the inherited SIMD scheme should not be used quantitatively in the manuscript until they are re-measured with the controlled integrator. Existing action-profile claims may be retained subject to their separate validated gates.

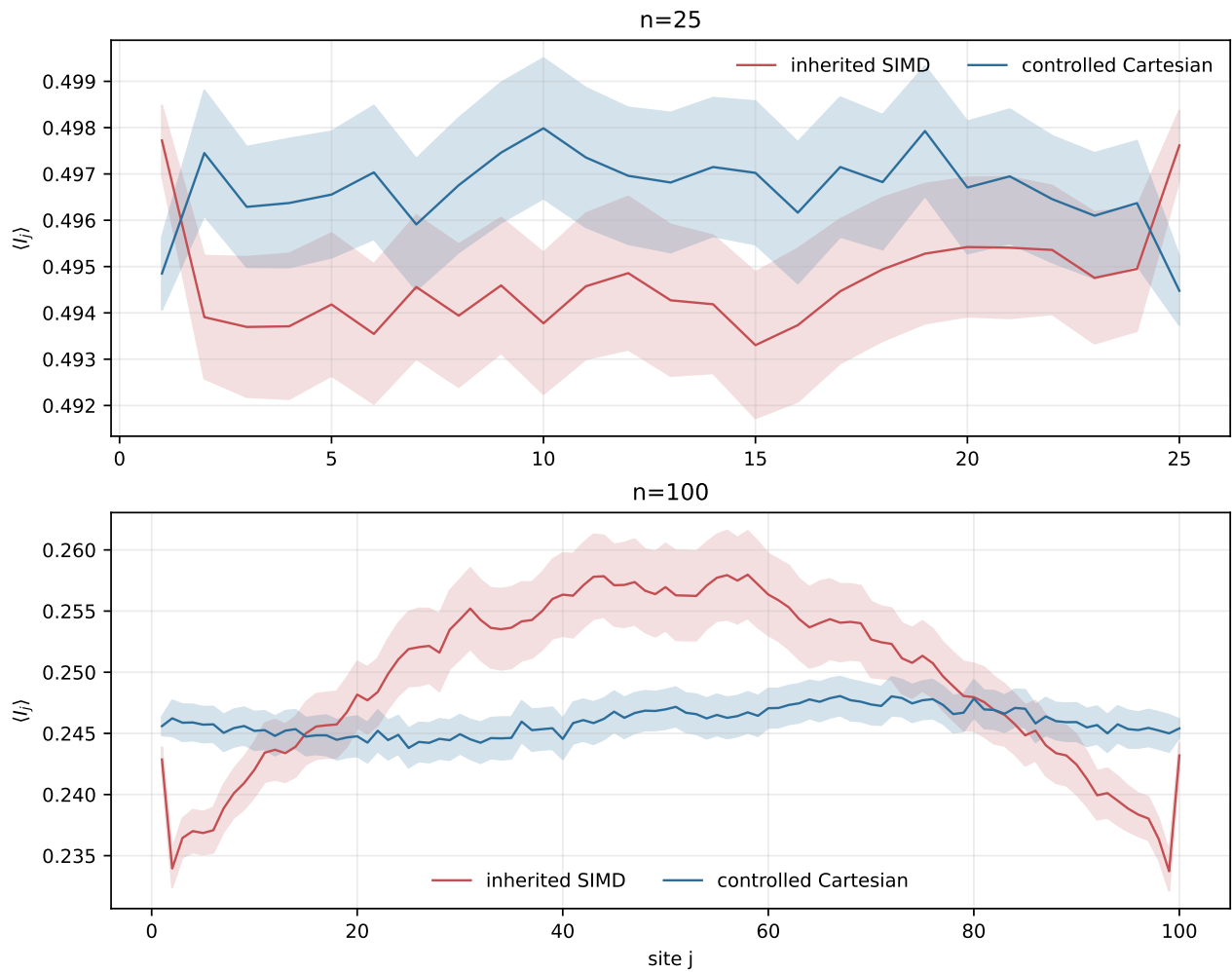


Figure 3: Equal-temperature action profiles. The controlled port preserves the action scale while eliminating the systematic bond-sine asymmetry.

8 Provenance

Launch commit: 908a7c08faac7ea3b8eb06c0577fca7fb5e9ad5e.

Remote frozen commit: 35124a825a4119e9007c635e739ae71e3752d6b7.

Source SHA-256: 17d2b0e75c7838c7fc8710bae0f0381c7607c967d38e03b4db3bde34e8ec086b.

Binary SHA-256: 68c13863da721f655ecbd3d0913b80691c0a0a3a6c984877af8423c331c443db. The 64 unique completed production commands are preserved in the provenance directory as `COMMANDS_UNIQUE_PRODUCTION.csv`. The original `COMMANDS.csv` retains all 72 launch records, including eight commands from the first detached attempt; that attempt produced no scientific data and its empty logs are preserved under `failed_detached_launch_1`.

A All bond-sine values

A.1 $n = 25$

j	ctrl. mean	ctrl. SE	ctrl. z	SIMD mean	SIMD SE	SIMD z
1	6.15916e-05	0.00023579	0.26121	0.00174083	0.000228514	7.6181
2	-0.000300712	0.000460072	-0.65362	0.00238436	0.000469214	5.0816
3	-9.25944e-05	0.000484063	-0.19129	0.00258475	0.00047951	5.3904
4	-0.000138459	0.000463517	-0.29871	0.00230554	0.000476456	4.8389
5	0.000557895	0.000479749	1.1629	0.00226579	0.000478774	4.7325
6	-4.94622e-05	0.00046166	-0.10714	0.0021442	0.000473771	4.5258
7	0.000728997	0.000454126	1.6053	0.00248401	0.00049485	5.0197
8	0.000796414	0.000453367	1.7567	0.00230344	0.000481128	4.7876
9	7.73374e-05	0.000473886	0.1632	0.0024502	0.000489438	5.0062
10	-6.38396e-05	0.000500351	-0.12759	0.00176292	0.000491034	3.5902
11	-0.000232355	0.000474126	-0.49007	0.000788587	0.000468305	1.6839
12	-0.000132196	0.000495086	-0.26702	0.00088276	0.000472852	1.8669
13	9.77672e-05	0.000482694	0.20254	0.000628334	0.000493684	1.2727
14	4.6572e-05	0.00046582	0.099979	0.000667911	0.000478831	1.3949
15	-0.000231863	0.000449133	-0.51625	8.83032e-05	0.000476094	0.18547
16	0.000855828	0.000456366	1.8753	-0.00114343	0.000483543	-2.3647
17	2.56119e-05	0.000463101	0.055305	-0.00151341	0.000480448	-3.15
18	0.000349486	0.000461511	0.75727	-0.00107992	0.000472016	-2.2879
19	0.000284919	0.000463956	0.61411	-0.00145148	0.000471163	-3.0806
20	-4.80647e-05	0.000479285	-0.10028	-0.00197725	0.000484072	-4.0846
21	-0.000319543	0.00046875	-0.68169	-0.002136	0.000476465	-4.483
22	0.000368907	0.000491523	0.75054	-0.00304327	0.00048648	-6.2557
23	0.000837944	0.00046049	1.8197	-0.00301796	0.000477369	-6.3221
24	0.000288455	0.000238637	1.2088	-0.00204457	0.0002356	-8.6781

A.2 $n = 100$

j	ctrl. mean	ctrl. SE	ctrl. z	SIMD mean	SIMD SE	SIMD z
1	0.000117568	0.000193383	0.60795	0.00252443	0.00019526	12.929
2	0.000190557	0.000621741	0.30649	0.00423771	0.000655634	6.4635
3	-0.000579927	0.000618558	-0.93755	0.00505453	0.000654098	7.7275
4	-0.000364176	0.000622344	-0.58517	0.00541257	0.00066381	8.1538
5	0.000728779	0.000643359	1.1328	0.00547403	0.000649299	8.4307
6	0.000776379	0.000624498	1.2432	0.00499207	0.000689233	7.2429
7	0.000172189	0.000672839	0.25591	0.00454461	0.000660658	6.8789
8	0.000387887	0.000674818	0.5748	0.00396825	0.000711925	5.574
9	0.000712575	0.000662449	1.0757	0.00469443	0.000690576	6.7978
10	-0.000489026	0.000651363	-0.75077	0.00472698	0.000687037	6.8802
11	0.000170656	0.000655408	0.26038	0.00426106	0.00070687	6.0281
12	0.0010045	0.000630471	1.5933	0.00513069	0.00069719	7.3591
13	0.000716707	0.000645378	1.1105	0.00472077	0.000672774	7.0169
14	-0.000165989	0.000659296	-0.25177	0.00495506	0.000685596	7.2274
15	-3.80497e-05	0.000672127	-0.056611	0.004537	0.000708173	6.4066
16	0.000512246	0.000680137	0.75315	0.00458986	0.000718993	6.3837
17	-0.00147243	0.000634706	-2.3199	0.0035845	0.000723534	4.9542
18	-0.00123338	0.000688561	-1.7912	0.0037957	0.000732921	5.1789
19	0.000327251	0.00069286	0.47232	0.0030392	0.000720766	4.2166
20	-5.31348e-05	0.000660133	-0.080491	0.00277851	0.000725791	3.8282
21	0.000482462	0.000672692	0.71721	0.00216696	0.000712798	3.0401
22	0.000881731	0.000667885	1.3202	0.003832	0.000777342	4.9296
23	0.000633444	0.000673031	0.94118	0.00362704	0.000704465	5.1486
24	-0.000389442	0.000651547	-0.59772	0.00319663	0.000705255	4.5326
25	-0.000528584	0.000655713	-0.80612	0.00202013	0.000688746	2.9331
26	0.000227436	0.000671807	0.33854	0.00237325	0.000682061	3.4795
27	-0.000812819	0.000642493	-1.2651	0.00359058	0.000714531	5.0251
28	-0.000172894	0.000661013	-0.26156	0.0036809	0.000713195	5.1611
29	0.00130636	0.000665555	1.9628	0.00253569	0.000714524	3.5488
30	0.000276185	0.000651829	0.42371	0.00259444	0.000702842	3.6913
31	0.000322316	0.00064058	0.50316	0.00113833	0.000704793	1.6151
32	0.000425434	0.000635262	0.6697	0.00215534	0.0006841	3.1506
33	-0.000212748	0.00062014	-0.34306	0.00209149	0.000738228	2.8331
34	-0.00038509	0.000688476	-0.55934	0.00157067	0.00068372	2.2972
35	-0.000841117	0.000625408	-1.3449	0.00206326	0.000682449	3.0233
36	0.000297027	0.000623805	0.47615	0.00105026	0.000735386	1.4282
37	0.000197912	0.000615865	0.32136	0.000287079	0.00069118	0.41535
38	1.21658e-05	0.000680466	0.017879	0.00122992	0.000689357	1.7842
39	-0.000483625	0.000666368	-0.72576	0.0011479	0.000681918	1.6833
40	0.000515675	0.000675597	0.76329	0.000683053	0.000696702	0.98041
41	-0.000380527	0.000643252	-0.59157	0.000600037	0.000704547	0.85166
42	-0.000369708	0.000616754	-0.59944	0.00158496	0.000743578	2.1315
43	-0.00092201	0.000667803	-1.3807	0.00133733	0.00068344	1.9568
44	-6.86376e-05	0.000649934	-0.10561	0.000242726	0.000713261	0.3403

j	ctrl. mean	ctrl. SE	ctrl. z	SIMD mean	SIMD SE	SIMD z
45	0.000301071	0.000617322	0.48771	0.000816815	0.000691317	1.1815
46	0.00100712	0.000644246	1.5633	0.00115351	0.000695826	1.6578
47	0.000937978	0.000673311	1.3931	0.000982132	0.000717265	1.3693
48	0.000687269	0.000656111	1.0475	0.000884114	0.000732792	1.2065
49	0.000230016	0.00066138	0.34778	0.00094038	0.000700969	1.3415
50	0.000580777	0.000646192	0.89877	0.00085233	0.000702686	1.213
51	0.000179482	0.000658228	0.27268	-0.000617296	0.000703657	-0.87727
52	0.000150586	0.000671548	0.22424	0.000488491	0.000677706	0.7208
53	-0.000317661	0.000636594	-0.499	-0.000162594	0.000681147	-0.23871
54	-0.000628833	0.000649377	-0.96836	-0.000342281	0.000699912	-0.48903
55	8.9059e-05	0.000622296	0.14311	0.000385899	0.000706267	0.54639
56	0.000346581	0.000633688	0.54693	-0.000414589	0.000679654	-0.61
57	-3.48977e-05	0.000646388	-0.053989	-0.000668478	0.000675302	-0.9899
58	-0.000344593	0.00063773	-0.54034	-0.000953963	0.000718607	-1.3275
59	0.000505914	0.000608927	0.83083	0.000191546	0.000703281	0.27236
60	-0.000114545	0.000679867	-0.16848	-0.000780364	0.000705091	-1.1068
61	-0.000500066	0.00065297	-0.76583	-0.00138258	0.000709662	-1.9482
62	0.000591755	0.000661437	0.89465	-0.00107066	0.00071652	-1.4942
63	-0.000586066	0.000656244	-0.89306	-0.00111601	0.00070029	-1.5936
64	-0.000167381	0.000638575	-0.26212	-0.00251716	0.000734295	-3.428
65	0.00125866	0.000617364	2.0388	-0.00210191	0.000690059	-3.046
66	0.000378213	0.000655416	0.57706	-0.00153359	0.000718302	-2.135
67	0.00101519	0.000646203	1.571	-0.00186727	0.000711234	-2.6254
68	-0.000378219	0.000654799	-0.57761	-0.00190585	0.000656278	-2.904
69	0.000137089	0.000646653	0.212	-0.00198338	0.000667741	-2.9703
70	-0.000683677	0.00067842	-1.0077	-0.00313011	0.000677459	-4.6204
71	0.000157577	0.000636902	0.24741	-0.00175454	0.000694355	-2.5269
72	0.000686884	0.000633077	1.085	-0.00232683	0.000685714	-3.3933
73	0.000727343	0.000654107	1.112	-0.00243385	0.000706558	-3.4446
74	-0.000861578	0.000626894	-1.3744	-0.00172796	0.000674	-2.5637
75	-9.30401e-05	0.000663201	-0.14029	-0.0027519	0.000686047	-4.0112
76	0.000433919	0.000657679	0.65977	-0.00348594	0.000704613	-4.9473
77	0.000252005	0.000648226	0.38876	-0.00338458	0.000690821	-4.8994
78	0.000439137	0.000651315	0.67423	-0.0022334	0.000723739	-3.0859
79	0.000371229	0.000670488	0.55367	-0.00228106	0.000689895	-3.3064
80	0.000983498	0.000634222	1.5507	-0.00261721	0.000681222	-3.8419
81	0.000985679	0.000650189	1.516	-0.00381041	0.000729565	-5.2229
82	0.00094879	0.000631328	1.5028	-0.00458439	0.000704097	-6.511
83	-0.000224551	0.000661633	-0.33939	-0.00345404	0.000726089	-4.7571
84	-0.00192651	0.000645606	-2.984	-0.00358658	0.000713924	-5.0238
85	-0.000460093	0.000647649	-0.7104	-0.00402858	0.000688539	-5.8509
86	-0.000762082	0.000619863	-1.2294	-0.00530198	0.000698177	-7.594
87	-0.00128713	0.00064364	-1.9998	-0.0046297	0.000750655	-6.1675
88	-0.000784828	0.000591597	-1.3266	-0.00540516	0.000726234	-7.4427
89	-9.42562e-05	0.000642759	-0.14664	-0.00528334	0.000698217	-7.5669
90	-0.00105221	0.000607964	-1.7307	-0.00547441	0.000684336	-7.9996
91	-0.000460395	0.000640498	-0.71881	-0.00529657	0.000646573	-8.1917
92	0.000197691	0.000620125	0.31879	-0.00427971	0.000678497	-6.3076
93	0.000973932	0.000655188	1.4865	-0.00439924	0.000673234	-6.5345
94	0.000451858	0.000639976	0.70605	-0.00486309	0.000668534	-7.2743
95	0.00106967	0.000659678	1.6215	-0.00638756	0.000670078	-9.5326
96	-0.000260969	0.000623204	-0.41875	-0.00619381	0.000686272	-9.0253
97	-0.0010299	0.000613709	-1.6782	-0.00615973	0.000651003	-9.4619
98	-0.000617765	0.000616082	-1.0027	-0.00469157	0.000649136	-7.2274
99	-0.000116681	0.000192011	-0.60768	-0.00222266	0.000203852	-10.903

B Full action profiles

B.1 $n = 25$

j	ctrl. mean	ctrl. SE	SIMD mean	SIMD SE	difference
1	0.494851	0.00039856	0.497723	0.000385703	-0.00287251
2	0.497451	0.000688705	0.493908	0.000676646	0.00354322
3	0.496289	0.000662098	0.493696	0.000771049	0.00259274
4	0.496372	0.000708227	0.49371	0.00080127	0.00266255
5	0.496555	0.000694966	0.494181	0.000784706	0.00237355
6	0.497034	0.000734785	0.493545	0.00076985	0.0034886
7	0.49591	0.000723452	0.494559	0.000791412	0.00135146
8	0.496757	0.000739303	0.49394	0.000786145	0.00281646
9	0.497461	0.000771285	0.494593	0.000745615	0.00286732
10	0.497986	0.000772239	0.493775	0.000779022	0.00421113
11	0.497356	0.000767663	0.494574	0.000803053	0.00278162
12	0.49696	0.000751841	0.494858	0.000842878	0.00210182
13	0.496815	0.000769919	0.494271	0.000833112	0.00254406
14	0.49715	0.000762312	0.494186	0.000758871	0.00296353
15	0.497024	0.000788579	0.493303	0.000804662	0.00372096
16	0.496164	0.000778576	0.493735	0.000843669	0.002429
17	0.49715	0.000768223	0.494467	0.000794466	0.00268287
18	0.496824	0.000741647	0.494941	0.000789197	0.00188295
19	0.497927	0.000713704	0.495279	0.000769403	0.00264835
20	0.496708	0.000727566	0.495421	0.000765069	0.00128641
21	0.496948	0.000738684	0.495408	0.000777104	0.0015407
22	0.496456	0.000696039	0.495359	0.00070798	0.00109697
23	0.496099	0.000690713	0.494756	0.000723627	0.00134282
24	0.49637	0.000685763	0.494949	0.000683413	0.00142065
25	0.494477	0.000377816	0.497615	0.000383806	-0.00313807

B.2 $n = 100$

j	ctrl. mean	ctrl. SE	SIMD mean	SIMD SE	difference
1	0.245591	0.000384898	0.242864	0.00051366	0.00272724
2	0.246234	0.000743112	0.233955	0.000793743	0.012279
3	0.245871	0.000768108	0.236451	0.00081271	0.00942019
4	0.245893	0.000747673	0.237009	0.000885779	0.00888423
5	0.245714	0.000739662	0.236853	0.000885459	0.00886044
6	0.245741	0.000735928	0.237079	0.000941077	0.00866176
7	0.245043	0.000764052	0.238866	0.000983273	0.0061765
8	0.245433	0.000778844	0.240107	0.00101802	0.0053259
9	0.245602	0.00077765	0.24094	0.00109773	0.00466192
10	0.245205	0.000797061	0.242026	0.00114398	0.00317876
11	0.245266	0.000799172	0.243421	0.00122329	0.00184472
12	0.244792	0.000817969	0.243661	0.00125013	0.00113173
13	0.245222	0.000820276	0.243382	0.00120569	0.00183973
14	0.245346	0.000802805	0.243907	0.00125345	0.00143902
15	0.244739	0.000783658	0.244997	0.00129757	-0.000257423
16	0.244841	0.000791479	0.245561	0.00135769	-0.00072039
17	0.244842	0.000835804	0.245662	0.00136355	-0.000820182
18	0.244475	0.000829075	0.245726	0.00137333	-0.00125176
19	0.244667	0.00082716	0.246744	0.00143368	-0.00207663
20	0.244765	0.000835011	0.248172	0.00137957	-0.00340751
21	0.244244	0.000829844	0.247702	0.00138695	-0.00345847
22	0.245214	0.000857684	0.248387	0.00138951	-0.00317378
23	0.244452	0.000812871	0.249859	0.00145652	-0.0054078
24	0.244879	0.000797714	0.251052	0.00150613	-0.00617363
25	0.243806	0.000814215	0.251893	0.00155246	-0.00808693
26	0.2443	0.00080605	0.252053	0.00161005	-0.00775311
27	0.244224	0.000819625	0.252154	0.0015493	-0.00792919
28	0.244554	0.000830094	0.251602	0.00160974	-0.00704782
29	0.244443	0.000827385	0.253476	0.00162879	-0.00903287
30	0.244936	0.000810498	0.25432	0.00165933	-0.00938373
31	0.244515	0.000821231	0.255206	0.00168757	-0.010691
32	0.244235	0.000852017	0.254279	0.00167128	-0.0100441
33	0.244613	0.00084461	0.253626	0.00166095	-0.0090134
34	0.244591	0.000884093	0.253518	0.00168594	-0.00892668
35	0.244628	0.000865443	0.25364	0.00169136	-0.00901202
36	0.245962	0.000855924	0.254159	0.00163474	-0.00819623
37	0.245265	0.000867089	0.25427	0.0015816	-0.0090043
38	0.245344	0.000860437	0.255002	0.00161408	-0.00965865
39	0.24542	0.000867503	0.255991	0.00165187	-0.0105709
40	0.244542	0.000843892	0.256344	0.00172423	-0.0118022
41	0.24584	0.000859357	0.256258	0.00174581	-0.0104174
42	0.246086	0.000866315	0.257107	0.00176384	-0.0110218
43	0.245842	0.000879005	0.257808	0.00176468	-0.0119658
44	0.246186	0.000857395	0.257852	0.00169924	-0.0116656

j	ctrl. mean	ctrl. SE	SIMD mean	SIMD SE	difference
45	0.246775	0.000850343	0.257112	0.00169358	-0.0103372
46	0.246271	0.000850042	0.25714	0.00176151	-0.0108691
47	0.246664	0.000837398	0.257372	0.00177978	-0.0107077
48	0.246856	0.000821391	0.256671	0.00173245	-0.00981456
49	0.246831	0.000804652	0.256383	0.00177066	-0.0095518
50	0.246962	0.000816301	0.256962	0.00181343	-0.010001
51	0.247171	0.000818389	0.256287	0.0018637	-0.00911631
52	0.246684	0.000813361	0.256265	0.00187104	-0.00958139
53	0.246571	0.000828858	0.256238	0.00185207	-0.00966723
54	0.246223	0.000838815	0.257083	0.00180099	-0.0108596
55	0.246509	0.000852541	0.257727	0.00180411	-0.0112185
56	0.246271	0.000852303	0.25794	0.00185795	-0.0116689
57	0.246412	0.000845726	0.257492	0.0017903	-0.0110793
58	0.24672	0.000847968	0.257978	0.00181781	-0.0112583
59	0.246434	0.000840399	0.257168	0.00179479	-0.0107341
60	0.247059	0.000812533	0.256351	0.00170143	-0.00929281
61	0.247076	0.000812529	0.255882	0.00170103	-0.00880573
62	0.247325	0.000803456	0.255303	0.0017453	-0.00797854
63	0.247458	0.000809496	0.254347	0.0016827	-0.00688916
64	0.247778	0.000791127	0.253665	0.00168362	-0.00588719
65	0.247582	0.000818681	0.254019	0.00174169	-0.0064367
66	0.247879	0.000814823	0.25434	0.00166599	-0.00646066
67	0.248051	0.000804635	0.254049	0.00161504	-0.00599787
68	0.247713	0.000846496	0.254117	0.00157771	-0.00640385
69	0.247591	0.000818257	0.254005	0.00156399	-0.00641415
70	0.247362	0.000831575	0.252677	0.00157658	-0.00531534
71	0.247244	0.000793397	0.252446	0.00150027	-0.00520186
72	0.248029	0.000804977	0.252315	0.00148956	-0.00428674
73	0.24789	0.000764393	0.251145	0.00146835	-0.00325507
74	0.247451	0.000780318	0.250766	0.00146245	-0.00331539
75	0.247706	0.000792669	0.251344	0.00145329	-0.00363822
76	0.247799	0.000809247	0.250743	0.00143293	-0.00294437
77	0.247326	0.000775163	0.249665	0.00141873	-0.00233987
78	0.246579	0.000770876	0.248847	0.00141902	-0.00226796
79	0.246691	0.000775043	0.248058	0.00143229	-0.00136677
80	0.24784	0.000781466	0.247958	0.00140661	-0.000117306
81	0.246961	0.00078699	0.247562	0.00144617	-0.00060143
82	0.246912	0.000798401	0.246889	0.00139643	2.33183e-05
83	0.246609	0.000764735	0.246511	0.00128781	9.79083e-05
84	0.247081	0.000768839	0.245741	0.00125995	0.00134057
85	0.247014	0.000785425	0.244838	0.00126343	0.00217653
86	0.245811	0.00077731	0.245223	0.00125223	0.000587729
87	0.246389	0.00078037	0.244027	0.0012621	0.00236213
88	0.246	0.000764941	0.243381	0.00124857	0.00261963
89	0.245914	0.000786432	0.243193	0.00124649	0.00272094
90	0.245925	0.00079273	0.242444	0.00124597	0.00348005
91	0.245486	0.000784579	0.241279	0.00113532	0.00420712
92	0.245682	0.000779679	0.239937	0.00101715	0.00574439
93	0.245001	0.000777971	0.240117	0.00102936	0.00488387
94	0.245743	0.00075825	0.239495	0.0010183	0.00624805
95	0.245352	0.000776831	0.238852	0.000947308	0.00649981
96	0.245272	0.000745491	0.238357	0.000894351	0.00691497
97	0.245447	0.00076356	0.238032	0.000901977	0.00741563
98	0.245233	0.000767487	0.236343	0.000856787	0.00889003
99	0.244999	0.000770357	0.233726	0.000806154	0.0112737
100	0.245406	0.000385367	0.243197	0.000510941	0.00220913